(6) (i) $\frac{1}{6}$ (ii) $\frac{17}{36}$ (iii) $\frac{5}{6}$

(7) (a)

x	0	1	2	3	4
$f(x)$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

(b) $\frac{4}{5}$ (c) $\frac{2}{5}$

EXERCISE 11.3

(1) 4

(2) (i) 0.16 (ii) 0.3 (iii) 0.75

(3) (i) $\frac{1}{400}$

(ii) $F(x)=\begin{cases} 0 & \text{for} \quad x<200 \\ \frac{x}{400}-\frac{1}{2} & \text{for} \quad 200\le x\le 600 \\ 1 & \text{for} \quad x>600 \end{cases}$

(iii) $\frac{1}{2}$

(4) (i) $\frac{1}{3}$ (ii) $1-e^{-\frac{x}{3}}$ (iii) $1-e^{-1}$ (iv) $e^{-\frac{5}{3}}$ (v) $1-e^{-\frac{4}{3}}$

(5) (i) $F(x)=\begin{cases} 0 & x\le -1 \\ \frac{x^2}{2}+x+\frac{1}{2} & -1\le x<0 \\ -\frac{x^2}{2}+x+\frac{1}{2} & 0\le x<1 \\ 1 & 1\le x \end{cases}$ (ii) 0.75

(6) (i) $f(x)=F'(x)=\begin{cases} 0 & x<0 \\ \frac{1}{2}(2x+1) & 0\le x<1 \\ 0 & 1\le x \end{cases}$ (ii) 0.285

EXERCISE 11.4

(1) (i) 2.3, 2.81 (ii) 1.67, 0.56 (iii) $\frac{5}{3}, \frac{1}{18}$ (iv) 2,4

(2) $\frac{8}{7}$

x	0	1	2
$f(x)$	$\frac{1}{7}$	$\frac{4}{7}$	$\frac{2}{7}$

(3) 7, 16

(4) 2, 1

x	0	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{16}$

(5) 15 minutes (6) $\frac{1}{3}$ (7) $\frac{1}{2}, \frac{1}{8}$ (8) Loss ₹. 0.50

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